

and growth in the electronics industry, obsolescence rate has also increased. People are phasing out/replacing their IT, communication and consumer electronics equipment including white and brown goods as shown in Table II.

As per a GTZ-MAIT sponsored study conducted recently by IMRB, e-waste generated in India during 2007 was around 332,979 MT besides about 50,000 MT entering the country by way of imports. The reasons for generation of this large quantity of e-waste were unprecedented growth of the IT industry during the last decade, and the early product obsolescence due to continuous innovation. Thus the net effect is the e-waste turning into a fastest growing waste stream.

However, the total e-waste available in 2007 for recycling and refurbishing was 144,143 MT. Of this, only 19,000 MT of e-waste could be processed.

Components of e-waste management

The major components of e-waste management are:

1. e-waste collection, sorting and transportation
2. e-waste recycling; it involves dismantling, recovery of valuable resource, sale of dismantled parts and export of processed waste for precious metal recovery

The stakeholders, i.e., the people who can help in overcoming the challenges posed by e-waste, are:

1. Manufacturers
2. Users
3. Recyclers
4. Policy makers

e-waste concerns and challenges

1. Accurate figures not available for rapidly increasing e-waste volumes—generated domestically and by imports
2. Low level of awareness among manufacturers and consumers of the hazards of incorrect e-waste disposal
3. No accurate estimates of the

TABLE III
WEE Generating
Top Ten States

State	WEE (Tonnes)
Maharashtra	20270.59
Tamil Nadu	13486.24
Andhra Pradesh	12780.33
Uttar Pradesh	10381.11
West Bengal	10059.36
Delhi	9729.15
Karnataka	9118.74
Gujarat	8994.33
Madhya Pradesh	7800.62
Punjab	6958.46

TABLE IV
WEE Generating
Top Ten Cities

City	WEE (Tonnes)
Ahmedabad	3287.5
Bangalore	4648.4
Chennai	4132.2
Delhi	9730.3
Hyderabad	2833.5
Kolkata	4025.3
Mumbai	11017.1
Nagpur	1768.9
Pune	2584.2
Surat	1836.5

quantity of e-waste generated and recycled available in India

4. Major portion of e-waste is processed by the informal (unorganised) sector using rudimentary techniques such as acid leaching and open-air burning, which results in severe environmental damage

5. e-waste workers have little or no knowledge of toxins in e-waste and are exposed to health hazards

6. High-risk backyard recycling operations impact vulnerable social groups like women, children and migrant labourers

7. Inefficient recycling processes result in substantial losses of material value and resources

8. Cherry-picking by recyclers who recover precious metals (gold, platinum, silver, copper, etc) and improperly dispose of the rest, posing environmental hazards

9. No specific legislation for dealing

with e-waste at present

Status of e-waste initiatives

The Ministry of Environment & Forests (MoEF) of the government of India is responsible for environmental legislation and its control. The Central Pollution Control Board (CPCB), an autonomous body under the MoEF, plays an important role in drafting guidelines and advising the MoEF on policy matters regarding environmental issues. Historically, in 2001 in cooperation with MoEF, the German Technology Cooperation (GTZ) began work on hazardous waste management in India through the advisory services in environmental management. Subsequently, Swiss Federal Laboratories for Material Testing and Research (EMPA) started to implement its global programme 'Knowledge Partnerships in e-waste Recycling.'

Combining the knowledge and technical expertise of EMPA on e-waste management, coupled with the field experience of the Indo-German projects in managing hazardous waste in India, the Indo-German-Swiss e-waste initiative was born in 2004. The vision of this initiative is to establish a clean e-waste channel that is a:

1. Convenient collection and disposal system for large and small consumers to return all their e-waste safely
2. Voluntary system for modern and concerned producers to care for their product beyond its useful life
3. Financially secure system that makes environmentally and socially responsible e-waste recycling viable

The objectives of the initiative are:

1. Reduce the risks to the population and the pollution of the environment resulting from unsafe handling
2. Focus on knowledge transfer to and skills upgrade of all involved stakeholders through trainings and seminars
3. Target mainly the existing informal recyclers allowing for their maximum but safe participation in future